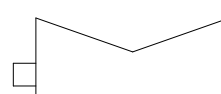
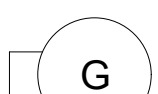
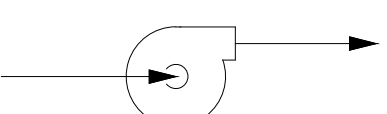
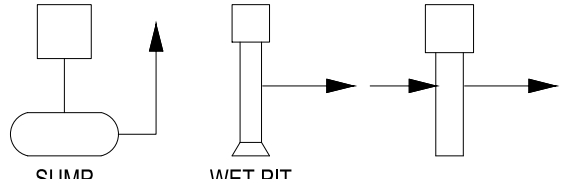
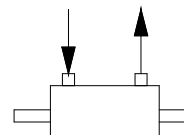
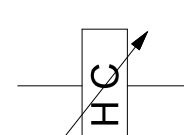
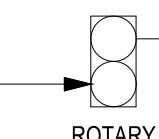
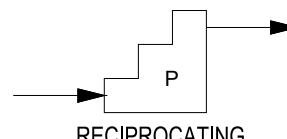
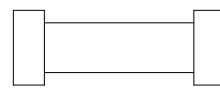
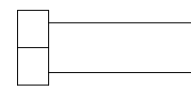
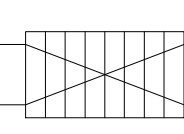
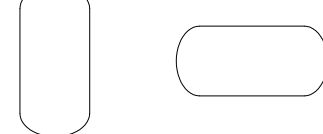
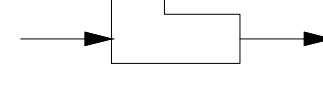
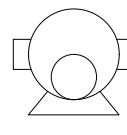
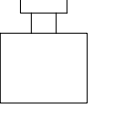
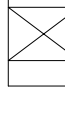
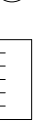
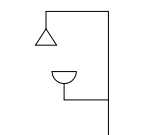
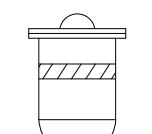
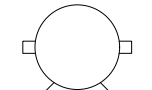
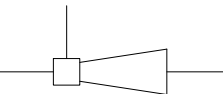
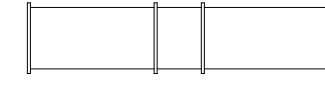
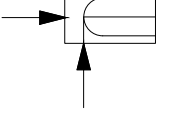
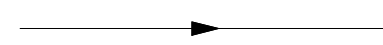
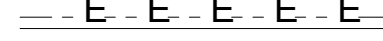
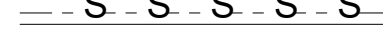
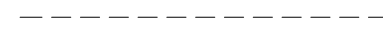
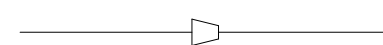
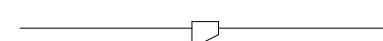
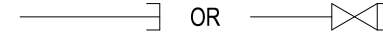
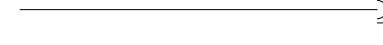
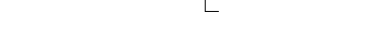
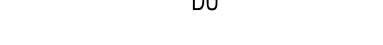
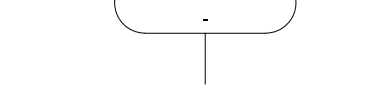
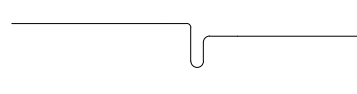
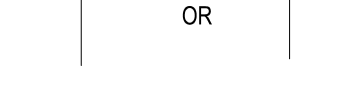
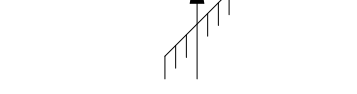
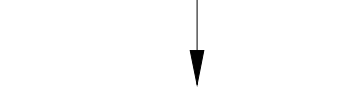
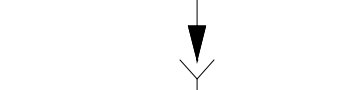
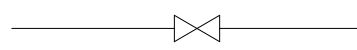
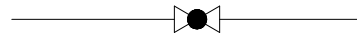
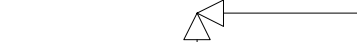
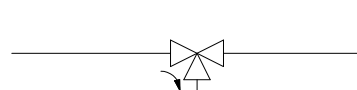
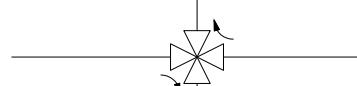
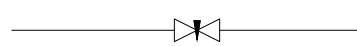
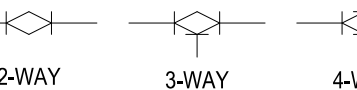
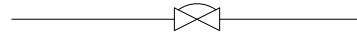
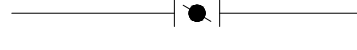
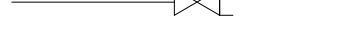
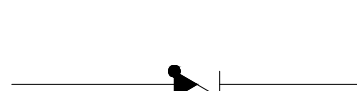
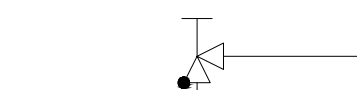
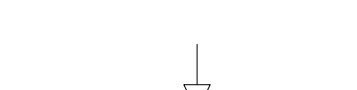
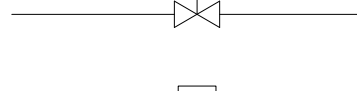
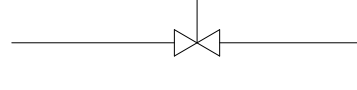
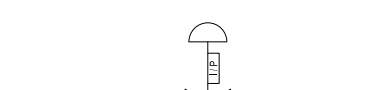
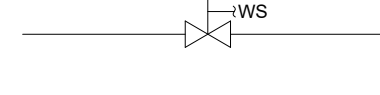
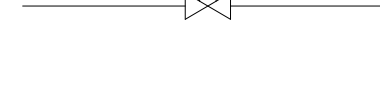
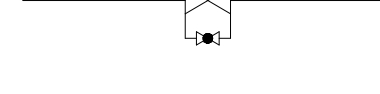
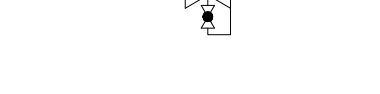
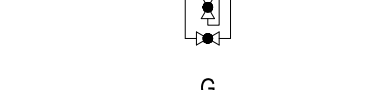
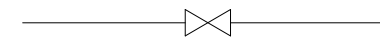
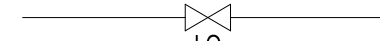
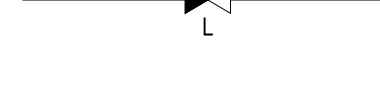
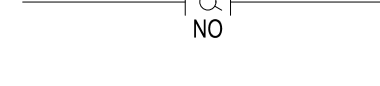
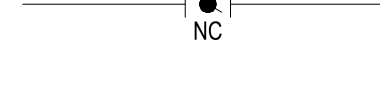
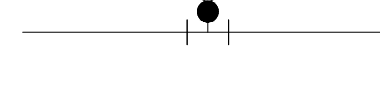
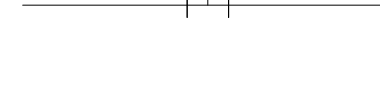
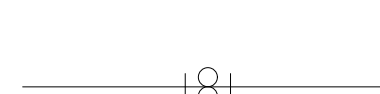
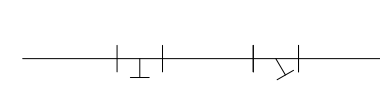
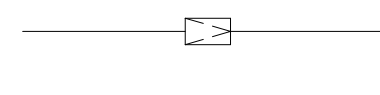
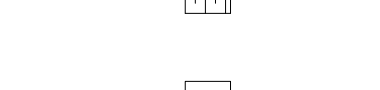
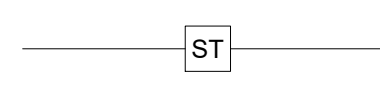
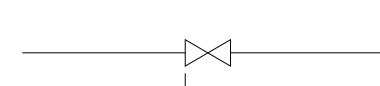
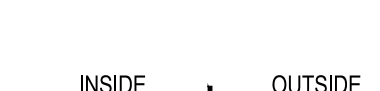
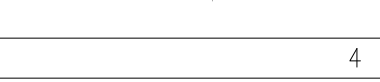
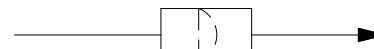
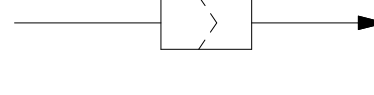
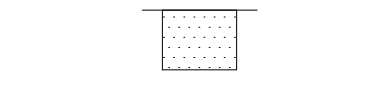
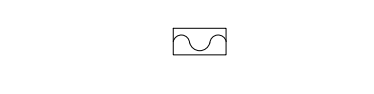
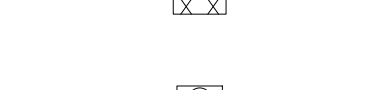
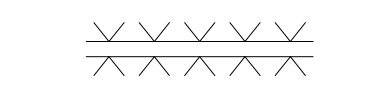
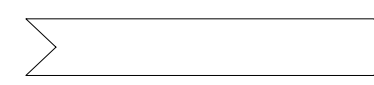
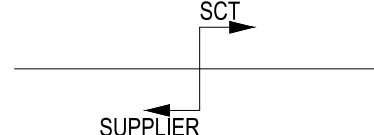
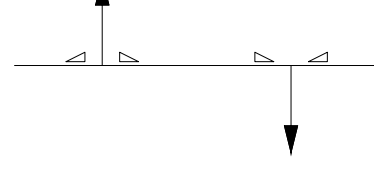
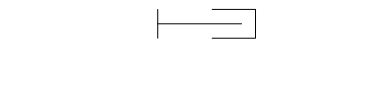
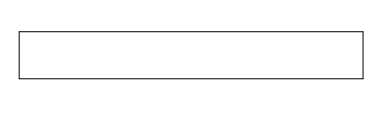
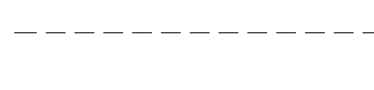
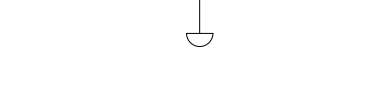
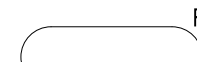
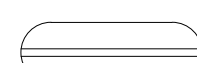
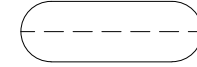
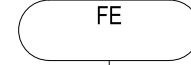
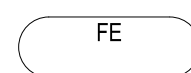
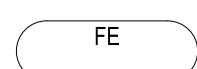
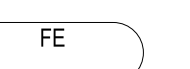
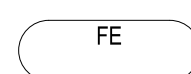
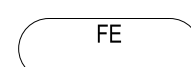
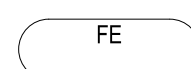
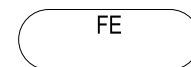
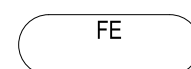
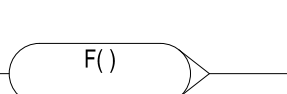
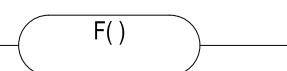
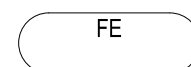
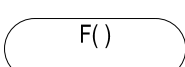
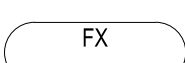
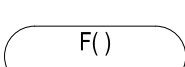
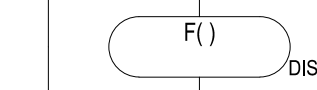
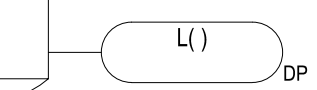
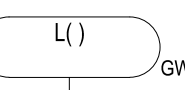
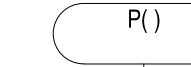
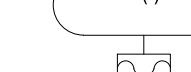
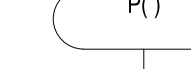
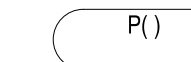
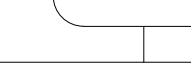
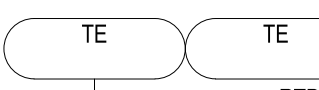
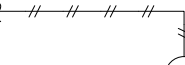
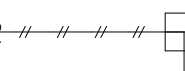
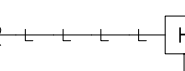
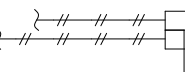
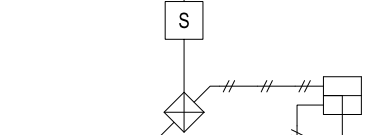
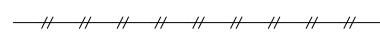
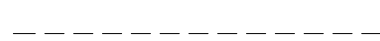
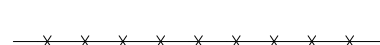
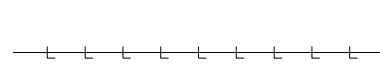
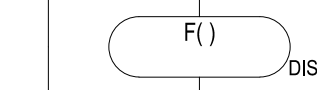
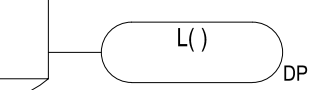
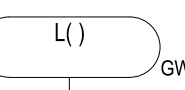
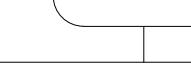
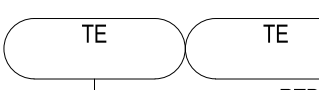
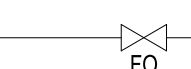
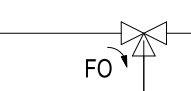


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EQUIPMENT		EQUIPMENT (CONT.)		PROCESS PIPING (CONT.)		LINE VALVES (CONT.)		MISCELLANEOUS (CONT.)		GENERAL NOTES :																																																																																																																											
 SINGLE FLOW  DOUBLE FLOW  GENERATOR  HORIZONTAL CENTRIFUGAL PUMP  VERTICAL PUMP  RING SECTION CENTRIFUGAL PUMP  VSD  HYDRAULIC COUPLING  ROTARY  RECIPROCATING  U-TUBE  STRAIGHT TUBE SINGLE-PASS  STRAIGHT TUBE TWO-PASS  PLATE TYPE HEAT EXCHANGER  PRESSURE VESSELS VERTICAL OR HORIZONTAL (TANK, RECEIVERS, DRYERS, SEPARATORS, ETC.)  CONE ROOF TANK  FLOATING ROOF TANK  RECIPROCATING AIR COMPRESSOR  ROTARY AIR COMPRESSOR  MIXER / AGITATOR  CHEMICAL STORAGE DRUM WITH DISPENSING PUMP  PRESSURIZED GAS BOTTLE  FUME SCRUBBER  CO2 ABSORBER  DAMPENER  CALIBRATION CYLINDER  EYE WASHER & SAFETY SHOWER  SAMPLE COOLER  VACUUM DEAERATOR  VACUUM PUMP  EJECTOR OR EDUCTOR  STEAM JET AIR EJECTOR COOLER  DESUPERHEATER PROCESS PIPING  MAIN PROCESS LINE  SECONDARY PROCESS LINE  HEAT TRACING E=ELECTRICALLY TRACED  S=STEAM TRACED  PIPING BY OTHER THAN SAMSUNG  BURIED PIPE PIPE LINE COMPONENT  CONCENTRIC REDUCER/ INCRERASER  ECCENTRIC REDUCER/ INCRERASER  WELDED CAP  PLUGGED PIPE OR VALVE  BREATHER CAP  HOSE CONNECTION  QUICK DISCONNECT  REMOVABLE SPOOL  FLEXIBLE JOINT  EXPANSION JOINT  BLIND FLANGE  INSULATING FLANGE (IF) OR DIELECTRIC UNION (DU) (SHOWN)  RO - RESTRICTION ORIFICE  LOOP SEAL  UPWARD & DOWNWARD SLOPE  VENT  EXHAUST VENT INTO THE OPEN AIR  DIRECT DRAIN ED : EQUIPMENT DRAIN CD : CHEMICAL DRAIN  INDIRECT DRAIN OR FUNNEL DRAIN LINE VALVES  GATE  GLOBE  ANGLE  THREE-WAY  FOUR-WAY  NEEDLE  PLUG  DIAPHRAGM  BALL  BUTTERFLY  HOSE  PINCH  CHECK  FOOT  FLOW LIMITING (OR EXCESS FLOW) CHECK  POWER ACTUATED NON-RETURN VALVE  BACKFLOW PREVENTER  STOP CHECK  ANGLE CHECK  TANDEM BLOW DOWN  MOTOR OPERATED VALVE  SOLENOID VALVE  CONTROL VALVE WITH POSITIONER  WATER SEAL  BELLOWS SEALED VALVE  PRESSURE BALANCING EXTERNAL PIPE WITH SMALL VALVE  PRESSURE RELIEVING EXTERNAL PIPE WITH SMALL VALVE  PRESSURE BALANCING & RELIEVING EXTERNAL PIPE WITH SMALL VALVE  ACTUATOR G = GEAR (SHOWN) A = AIR WRENCH C = CHAIN X = EXTENSION STEM B = WITH INTEGRAL BYPASS VALVE D = DEEP STUFFING BOX AND LANTERN GLAND T = TAP IN BOTTOM OF BODY (FOR ASME TEST) VALVE STATUS SYMBOLS  OPEN DURING NORMAL OPERATION (ALL VALVES EXCEPT BUTTERFLY)  LOCKED OPEN  CLOSED DURING NORMAL OPERATION (ALL VALVES EXCEPT BUTTERFLY)  LOCKED CLOSED  THROTTLE (USED WHEN LOCKED IN THROTTLE POSITION)  OPEN DURING NORMAL OPERATION (BUTTERFLY ONLY) (SUBSTITUTE LO FOR LOCKED OPEN)  CLOSED DURING NORMAL OPERATION (BUTTERFLY ONLY) (SUBSTITUTE LC FOR LOCKED CLOSED) MISCELLANEOUS  SPECTACLE BLIND  CIRCULAR OR HAMMER BLIND  SIMPLEX BASKET STRAINER  DUPLEX BASKET STRAINER  Y-TYPE / T-TYPE STRAINER (SHOWN WITHOUT BLOWOFF VALVE)  TEMPORARY STARTUP STRAINER  SCREEN TYPE STRAINER  FILTER  FLAME ARRESTER  TRAP ST = STEAM TRAP (SHOWN) AT = AIR TRAP DT = DRAIN TRAP RT = RESIN TRAP WT = WATER TRAP (AUTO AIR VENT)  PIPE CLASS CHANGE  BUILDING PENETRATION  DEBRIS FILTER  BALL SCREEN  BIRD SCREEN  VACUUM BREAKER  DIAPHRAGM SEAL  VORTEX BREAKER  FLOW GLASS  SEAL PIT  SPRAY NOZZLE OR SPARGER (SHOWN WITH UPWARD/DOWNWARD FLOW)  PROCESS LINE BREAK OR RESUMPTION POINT  SUPPLIER INTERFACE  PIPING HIGH POINT VENT (V) OR PIPING LOW POINT DRAIN (D)  SLIDING GATE  INSTRUMENTATION LINE BREAK OR RESUMPTION POINT  DESIGNATION FOR EQUIPMENT AND PIPING WHICH APPEAR ON OTHER P & I D'S  BOUNDARY OF EQUIPMENT PACKAGE PROVIDED BY SUPPLIER.  DRAIN LEG, DRIP POT GENERAL NOTES : 1. THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE. NOTES : FOR INTERNAL USE THIS DRAWING IS THE PROPERTY OF SAMSUNG C&T CORPORATION IT IS NOT TO BE COPIED OR USED IN ANY WAY DETRIMENTAL TO THE COMPANY. <table><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>A</td><td>19.SEP.2025</td><td>FOR INTERNAL USE</td><td></td><td></td><td></td></tr><tr><td>REV.</td><td>DATE</td><td>DESCRIPTION</td><td>PREPARED</td><td>REVIEWED</td><td>APPROVED</td></tr><tr><td>EMPLOYER</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td colspan="6">TENDERER</td></tr><tr><td colspan="6">PROJECT NAME</td></tr><tr><td colspan="6">DRAWING TITLE</td></tr><tr><td colspan="6">PROJECT DWG NO.</td></tr><tr><td colspan="6">PROJECT NO.</td></tr><tr><td colspan="6">REF. DRAWING NO.</td></tr><tr><td colspan="6">D00P</td></tr><tr><td colspan="4">SHEET</td><td colspan="2">REV.</td></tr><tr><td colspan="4">1/1</td><td colspan="2">A</td></tr><tr><td colspan="4">SCALE</td><td colspan="2">REF.CODE</td></tr><tr><td colspan="4">NONE</td><td colspan="2"></td></tr></table> AL NOUF1 PROJECT P&ID FOR SYMBOLS & LEGENDS 1 19.SEP.2025 FOR INTERNAL USE 1/1 A 1 2 3 4 5 6 7 8 A1 (594 X 841 MM)																																																		A	19.SEP.2025	FOR INTERNAL USE				REV.	DATE	DESCRIPTION	PREPARED	REVIEWED	APPROVED	EMPLOYER						TENDERER						PROJECT NAME						DRAWING TITLE						PROJECT DWG NO.						PROJECT NO.						REF. DRAWING NO.						D00P						SHEET				REV.		1/1				A		SCALE				REF.CODE		NONE					
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FIRST LETTER		SUCCEEDING LETTERS																		SELF-ACTUATED DEVICES-PRESSURE				GENERAL NOTES : 1. THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE.			
		SENSING DEVICE		DISPLAY DEVICE						CONTROL DEVICE				MISCELLANEOUS DEVICE													
SYMBOL	MEASURED OR INITIATING VARIABLE	PRIMARY ELEMENT	TRANSMITTER	INDICATOR	RECORDER	INTEGRATING INDICATOR	ARARM ANNUNCIATOR			CONTROL STATION MAN/AUTO	CONTROLLER	CONTROL VALVE	OTHER FINAL CONTROL ELEMENT	SWITCH	LOCAL OBSERVATION GLASS	SOLENOID RELAY OR POSITIONER	TEST POINT	USER'S CHOICE	UNCLASSIFIED	NOTES : PRESSURE RELIEF OR SAFETY VALVE ANGLE PATTERN SPRING OR WEIGHT LOADED OR WITH INTEGRAL PILOT VALVE OR WITH INTEGRAL PILOT VALVE VACUUM RELIEF VALVE ANGLE PATTERN SPRING OR WEIGHT LOADED OR WITH INTEGRAL PILOT VALVE PRESSURE RELIEF OR SAFETY VALVE BY INTEGRAL SOLENOID RUPTURE DISC OR SAFETY HEAD FOR PRESSURE RELIEF RUPTURE DISC OR SAFETY HEAD FOR VACUUM RELIEF							
()	TYPICAL SYMBOL	()E	()T	()I	()R	()QI	()AL	()AH	()AHL	()K	()C	()CV	()Z	()S	()G	()Y	()P	()N	()X								
A	ANALYSIS	A E	A T	A I	A R		AAL	AAH	AAHL	A K	A C		A Z	A S		A Y	A P	A N	A X								
B	BURNER FLAME	B E	B T	B I	B R		BAL				B C			B S		B Y	B P	B N	B X								
C	CONDUCTIVITY	C E	C T	C I	C R		CAL	CAH	CAHL	C K			C Z	C S		C Y	C P	C N	C X								
D	DENSITY (MASS) OR SPECIFIC GRAVITY	D E	D T	D I	D R		DAL	DAH	DAHL	D K	D C		D Z	D S		D Y	D P	D N	D X								
E	VOLTAGE (EMF)	E E	E T	E I	E R		EAL	EAH	EAHL	E K	E C		E Z	E S		E Y		E N	E X								
F	FLOW	F E	F T	F I	F R	FQI	FAL	FAH	FAHL	F K	F C	FCV	F Z	F S	FG	F Y	F P	F N	F X								
G	GAUGING (DIMENSIONAL)	G E	G T	G I	G R		GAL	GAH	GAHL	G K	G C			G S		G Y		G N									
H	HAND (MANUAL INITIATED)	H									H C		H Z	H S		H Y		H N									
I	CURRENT	I E	I T	I I	I R	IQI	IAL	IAH	IAHL	I K			I Z	I S		I Y		I N	I X								
J	POWER	J E	J T	J I	J R	JQI	JAL	JAH	JAHL	J K			J Z	J S		J Y		J N	J X								
K	TIME	K E	K T	K I	K R	KQI	KAL	KAH	KAHL		K C		K Z	K S		K Y		K N	K X								
L	LEVEL	L E	L T	L I	L R		LAL	LAH	LAHL	L K	L C	LCV	L Z	L S	LG	L Y	LP	LN	L X								
M	MOISTURE OR HUMIDITY	M E	M T	M I	M R		MAL	MAH	MAHL	M K	M C		M Z	M S		M Y	MP	M N	M X								
O	USER'S CHOICE																										
P	PRESSURE OR VACUUM	P E	P T	P I	P R		PAL	PAH	PAHL	P K	P C	PCV	P Z	P S		P Y	P P	P N	P X								
PD	PRESSURE DIFFERENTIAL	PDE	PDT	PDI	PDR		PDAL	PDAH	PDAHL	PDK	PDC		PDZ	PDS													
Q	QUANTITY	Q E	Q T	Q I	Q R	QQI	QAL	QAH	QAHL	Q K	Q C		Q Z	Q S		Q Y		Q N	Q X								
S	SPEED OR FREQUENCY	S E	S T	S I	S R	SQI	SAL	SAH	SAHL	S K	S C		S Z	S S		S Y		S N	S X								
T	TEMPERATURE	T E	T T	T I	T R		TAL	TAH	TAHL	T K	T C	TCV	T Z	T S		T Y	T P	T N	T X								
U	MULTIVARIABLE			U I	U R								U Z	U S		U Y		U N	U X								
V	VIBRATION		V T	V I	V R		VAL	VAH	VAHL	V K	V C		V Z	V S		V Y	V P	V N	V X								
W	WEIGHT OR FORCE	W E	W T	W I	W R	WQI	WAL	WAH	WAHL	W K	W C		W Z	W S		W Y		W N	W X								
X	UNCLASSIFIED																										
Y	EVENT OR STATE	Y E	Y T	Y I	Y R		YAL	YAH					Y Z	Y S		Y Y		Y N	Y X								
Z	POSITION	Z E	Z T	Z I	Z R		ZAL	ZAH	ZAHL				ZZ	Z S		Z Y		Z N	Z X								
GENERAL INSTRUMENTS				SENSOR-FLOW						SENSOR-FLOW (CONT.)				SENSORS-PRESSURE				VALVES ACTUATORS									
 FF INSTRUMENT FOR SINGLE MEASURED VARIABLE AND ANY NUMBER OF FUNCTIONS. INSTRUMENT IS LOCALLY MOUNTED.  FF INSTRUMENT FOR SINGLE MEASURED VARIABLE AND ANY NUMBER OF FUNCTIONS. INSTRUMENT IS LOCALLY MOUNTED. FF=FOUNDATION FIELDBUS  INSTRUMENT IS MOUNTED ON A LOCAL PANEL (TYPICAL)  INSTRUMENT IS MOUNTED ON A MAIN CONTROL PANEL (TYPICAL)  INSTRUMENT FOR PERFORMANCE TEST ONLY  INSTRUMENT FOR PERFORMANCE TEST PLUS NORMAL USE  INSTRUMENT IS MOUNTED BEHIND A PANEL (TYPICAL)				 FE FE - ORIFICE PLATE  FE ORIFICE PLATE IN QUICK - CHANGE FITTING  FE  FE VENTURI TUBE / FLOW NOZZLE  FE PITOT TUBE OR PITOT VENTURI TUBE  FE FORWARD-REVERSE PITOT TUBE  FE FLUME  FE VORTEX  FE TURBINE OR PROPELLER TYPE  F() ROTAMETER  F() IN LINE INSTRUMENT  FE						 F() ULTRASONIC  FX FLOW STRAIGHTENING VANES  F() POSITIVE DISPLACEMENT TYPE  F() DIS DIS - DISPLACEMENT (SHOWN) BNN - BNNNETIC DP - DIFFERENTIAL PRESSURE FLO - FLOAT REF - REFLEX FLAT GLASS  L() DP DP - DIFFERENTIAL PRESSURE (SHOWN) BNN - BNNNETIC  L() GWR GWR - GUIDED WAVE RADAR (SHOWN) RAD - RADAR FLO - FLOAT DIS - DISPLACEMENT				 P() DIRECT CONNECTED  P() WITH DIAPHRAGM SEAL PIPED  P() WITH DIAPHRAGM SEAL LINE OR EQUIP MOUNTED  P() STRAIN-GAGE TYPE  T() TW - THERMOWELL TE - TEMPERATURE ELEMENT WITH THERMOWELL TIT - TEMPERATURE TRANSMITTER WITH THERMOWELL AND TEMPERATURE ELEMENT  TE  TE RTD or K DUAL SENSOR (SHOWN WHEN BOTH ARE USED) RTD-RESISTANCE TEMPERATURE DETECTOR K- K type THERMO COUPLE  T() FILLED SYSTEM				 PNEUMATIC DIAPHRAGM  PNEUMATIC DIAPHRAGM WITH POSITIONER  DIAPHRAGM PRESSURE-BALANCED  ROTARY MOTOR (SHOWN TYPICALLY WITH ELECTRIC SIGNAL)  PNEUMATIC CYLINDER SINGLE-ACTING  HYDRAULIC CYLINDER SINGLE-ACTING  PNEUMATIC CYLINDER DOUBLE-ACTING  PNEUMATIC CYLINDER DOUBLE ACTING WITH PILOT VALVE  SINGLE SOLENOID (NOTE 14) RESET AFTER LATCH (OPTIONAL)  ELECTRO OR HYDRAULIC  HAND ACTUATOR (MOUNTED AT THE TOP SIDE OR BOTTOM OF ACTUATED DEVICE AS APPLICABLE)  UNCLASSIFIED (TYPE OF ACTUATOR TO BE WRITTEN ADJACENT TO THE SYMBOL)									
SIGNALS & LINES										SENSOR-LEVEL				SENSORS-TEMPERATURE				VALVE BODIES & FAILURE MODES									
 CONNECTION TO PROCESS OR MECHANICAL LINE OR INSTRUMENT SUPPLY  PNEUMATIC SIGNAL OR UNDEFINED SIGNAL FOR PROCESS FLOW DIAGRAMS  ELECTRIC SIGNAL  CAPILLARY TUBE (FILLED SYSTEM)  HYDRAULIC SIGNAL  ELECTROMAGNETIC OR SONIC SIGNAL  SOFTWARE OR DATA LINK										 F() DIS DIS - DISPLACEMENT (SHOWN) BNN - BNNNETIC DP - DIFFERENTIAL PRESSURE FLO - FLOAT REF - REFLEX FLAT GLASS  L() DP DP - DIFFERENTIAL PRESSURE (SHOWN) BNN - BNNNETIC  L() GWR GWR - GUIDED WAVE RADAR (SHOWN) RAD - RADAR FLO - FLOAT DIS - DISPLACEMENT				 T() TW - THERMOWELL TE - TEMPERATURE ELEMENT WITH THERMOWELL TIT - TEMPERATURE TRANSMITTER WITH THERMOWELL AND TEMPERATURE ELEMENT  TE  TE RTD or K DUAL SENSOR (SHOWN WHEN BOTH ARE USED) RTD-RESISTANCE TEMPERATURE DETECTOR K- K type THERMO COUPLE  T() FILLED SYSTEM				 FO TWO - WAY (TYPICAL GLOBE SHOWN) FO - INDICATES FAIL OPEN FC - INDICATES FAIL CLOSED FL - INDICATES LOCKED (POSITION DOES NOT CHANGE) FI - INDICATES FAIL INDETERMINATE  FO THREE - WAY									
				SENSOR-ANALYSIS																							
				 A() THROUGH-FLOW TYPE  A()																							
8		7			6			5			4			3			2			1							

8

7

6

5

4

3

2

1

TYPICAL IDENTIFICATION SYSTEM FOR
PIPE SIZE,
CLASS, MATERIAL AND INSULATION

EXAMPLE OF
APPLICATION :

A

A

2

A

A

N

(1)

(2)

(3)

(1) CLASS/RATING

A : 4500# or 2500#SP (ASME B16.5)

B : 2500# (ASME B16.5)

C : 1500#SP (ASME B16.5)

D : 1500# (ASME B16.5)

E : 900# (ASME B16.5)

F : 600# (ASME B16.5)

G : 300# (ASME B16.5)

H : 150# (ASME B16.5)

J : 125# (ASME B16.1)

X : Special as Designated

(2) PIPING MATERIAL

A : ALLOY STEEL (C-Mn)

B : CARBON STEEL (CS)

C : STAINLESS STEEL (304 or 304L)

D : STAINLESS STEEL (316 or 316L), DUPLEX STAINLESS STEEL

E : CAST IRON or DUCTILE IRON (CI or DI)

F : GALVANIZED CARBON STEEL

G : COPPER

H : PVC or CPVC

J : PVDF

K : PLASTIC LINED CARBON STEEL

L : HIGH DENSITY POLYETHYLENE (HDPE)

M : POLYPROPYLENE (PP)

R : FRP, GRP, GRE, GRV, etc.

S : CONCRETE

(3) SERIAL NO.

TYPICAL IDENTIFICATION SYSTEM FOR
PIPE LINE, VALVE, EQUIPMENT AND FILTER, SCREEN,STRAINER, etc.

TAG NUMBERING SYSTEM IS BASED ON THE KKS NUMBERING SYSTEM. (DOC. NO.: 19K9-00GEN00-G02-0002)

0

BREAKDOWN LEVEL 1

U

U

S

S

S

W

W

A

A

N

N

N

NUMERIC SEQUENCE IN EACH SYSTEM
(E.G. 00,01,02, etc.)

EQUIPMENT NUMERIC SEQUENCE

UNIT IDENTIFICATION NUMBERS

	PLANT UNIT	"UU" CHARACTER DESIGNATION
POWER PLANT	POWER PLANT COMMON	0
	FIRST GROUP COMMON (INCL. FIRST GROUP STG)	10
	FIRST GROUP #1 GTG / HRSG	11
	FIRST GROUP #2 GTG / HRSG	12
	SECOND GROUP COMMON (INCL. SECOND GROUP STG)	20
	SECOND GROUP #1 GTG / HRSG	21
	SECOND GROUP #2 GTG / HRSG	22

THE BASIS FOR THE DESIGNATION OF INSTALLATIONS AND SUBSYSTEMS
IN POWER PLANTS.

SERIAL NO. OF
BREAKDOWN LEVEL

DESIGNATION OF
BREAKDOWN LEVEL

0

UNIT
NUMBER

1

FUNCTION
CODE
(KKS)

SEQUEN-
TIAL
NUMBER

2

EQUIPMENT
UNIT
CODE

3

COMPONENT
CODE

EXAMPLE OF
APPLICATION:

20LAB31AP001 (COGENERATION FEEDWATER PUMP A)

0

00

1

LAB

01

2

AP001

3

COMPONENT
CODE

20LAB11AP001-M (COGENERATION FEEDWATER PUMP A MOTOR)

0

00

1

LAB

01

2

AP001

3

-M

PIPING DESIGNATION FLAG
(POINTING IN FLOW DIRECTION INCLUDING PIPING DIAMETER,
PROCESS DESIGN CODE, PIPING INSULATION)

11LAB11
BR001
250H LAB1

FLOW DIRECTION

BREAKDOWN LEVEL 0&1
BREAKDOWN LEVEL 2

PROCESS DESIGN CODE

PIPING INSULATION
- H : INSULATION FOR HEAT CONSERVATION
- P : INSULATION FOR PERSONAL PROTECTION
- N : NO INSULATION

PIPING DIAMETER

EXAMPLE FOR LINE VALVE, EQUIPMENT AND STRAINER IDENTIFICATION

00LAB11
BR001
250H LAB1

00LAB11
AA001

00LAB11
AT001

00LAB01
AP'001

00LAB21
BR001
250H LAB2

GENERAL NOTES :

1. THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE
ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL
DESIGN PHASE.

NOTES :

8

7

6

5

4

3

2

1

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A	19.SEP.2025	FOR INTERNAL USE			
REV.	DATE	DESCRIPTION	PREPARED	REVIEWED	APPROVED
EMPLOYER					
TENDERER					
PROJECT NAME					
AL NOUF1 PROJECT					
DRAWING TITLE					
P&ID FOR SYMBOLS & LEGENDS 4					
PROJECT DWG NO.				SHEET	REV.
D00P-00GEN00-M05-0005				1/1	A
PROJECT NO.	REF. DRAWING NO.			SCALE	REF.CODE
D00P				NONE	

NUMBER
RANGE

VALVES

PIPE BR

INSTRUMENTATION
CIRCUITS

001-100

-VALVES IN MAIN PIPES
-VALVE GENERAL

-MAIN PIPES

-TRANSMITTER

101-190

-CONTROL VALVES / DAMPERS

-SWITCH

191-199

-SAFETY VALVES

-SUCTION AND PRESSURE RELIEF
LINES ON SAFETY VALVES

201-250

-TCE IS USING FOR MOV IN MAIN
LINES

-NOT USED

-FOR TESTING EQUIPMENTS

251-299

-SHUT OFF VALVE MAIN LINE

301-399

-INSTRUMENT ISOLATION

-PRESSURE LINES FOR
INSTRUMENTS

-NOT USED

401-499

-VALVES IN DRAINS AND
FLUSHING PIPES

-DRAINS, FLUSHING PIPES,
PRESSURE SUPPERSSION PIPES

-ACCEPTANCE TESTS
-TEST THERMOWELL

501-599

-VALVES FOR VENTING

-VENTING

-LOCAL MEASURING CIRCUITS
-GAUGES

601-699

-VALVES FOR SAMPLING AND
DOSING

-SAMPLING AND DOSING PIPES

-FLOW ELEMENT

701-799

-VALVES FOR INTERNAL
CONTROLS

-FOR INTERNAL CONTROLS

-GROUPED

801-899

1. TCE IS USING FOR MOV IN DRAIN LINES
2. STEAM TRAP

-NOT USED

-GROUPED

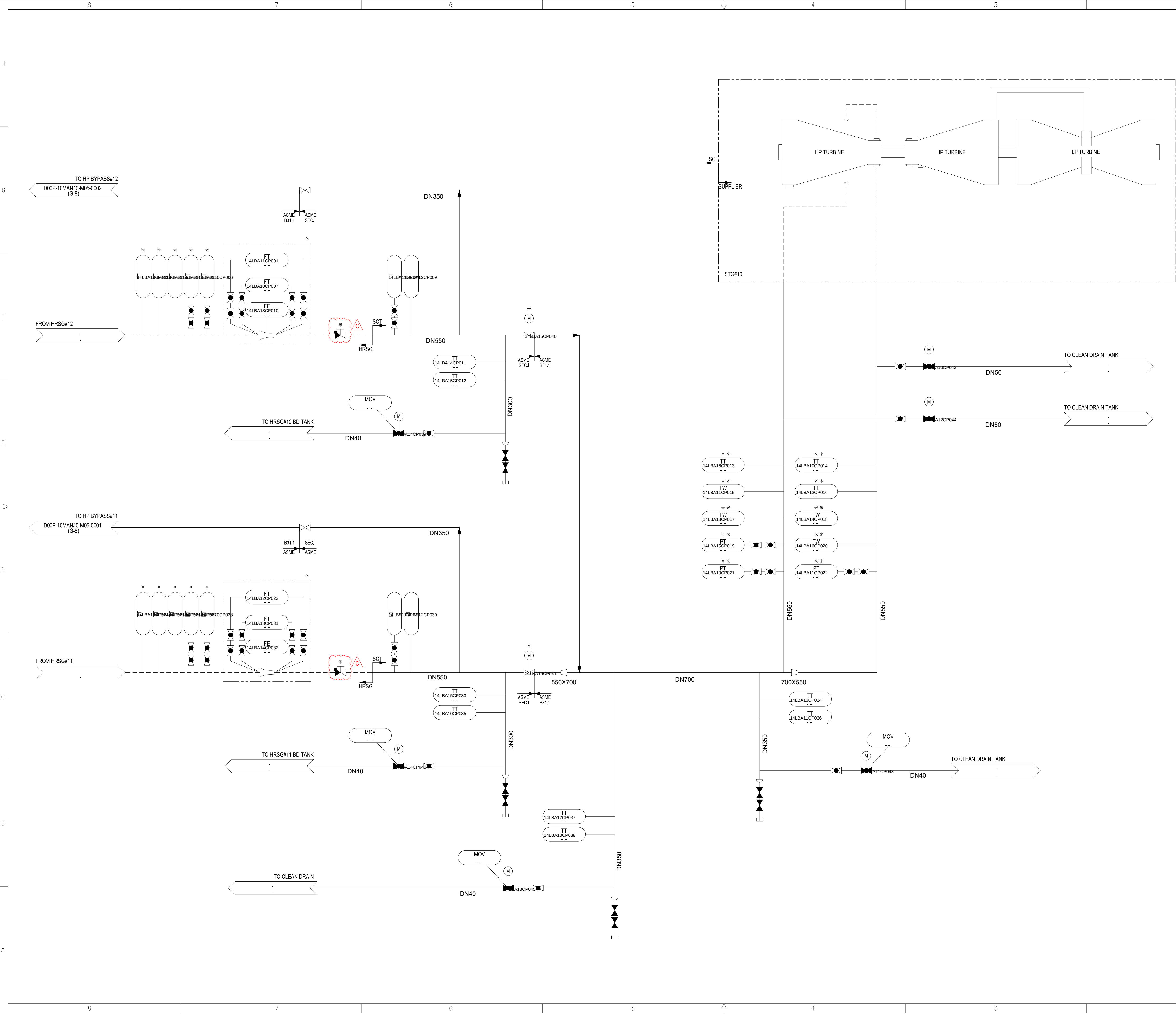
901-999

-TCE IS USING FOR MOV IN VENT
LINES

-NOT USED

-GROUP MEASURING CIRCUITS
(E.G. MULTI POINT SELECTOR
SWITCHES, MULTI CHANNEL
RECORDERS)

A1 (594 X 841 MM)



- GENERAL NOTES :
- REFER TO SYMBOL & LEGEND
DWG. NO. D00P-00GEN00-M05-0002-0005
 - THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE.
- NOTES :
- THIS P&ID IS FOR GROUP#10, CONFIGURATION IS IDENTICAL FOR GROUP#20.
 - DRAIN POT AND WARM-UP LINE SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM STEAM TURBINE MAIN STOP & CONTROL VALVE.
 - DRAIN VALVES SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM DRAIN POT.
- * DENOTES EQUIPMENT WILL BE SUPPLIED BY HRSG.
- ** DENOTES EQUIPMENT WILL BE SUPPLIED BY ST SUPPLIER.

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C	24.NOV.2025	FOR INTERNAL USE			
B	01.OCT.2025	FOR INTERNAL USE			
A	19.SEP.2025	FOR INTERNAL USE			
REV.	DATE	DESCRIPTION	PREPARED	REVIEWED	APPROVED
EMPLOYER					
TENDERER					
PROJECT NAME					
AL NOUF1 PROJECT					
DRAWING TITLE					
P&ID FOR HP STEAM SYSTEM GROUP 10					
PROJECT DWG NO.				SHEET	REV.
D00P-10LBA10-M05-0001				1/1	C
PROJECT NO.		REF. DRAWING NO.		SCALE	REF. CODE
D00P				NONE	

